

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$a. H_0: \sigma_M^2 = \sigma_F^2 \text{ v.s. } H_1: \sigma_M^2 \neq \sigma_F^2 \quad (H_0: \frac{\sigma_M^2}{\sigma_F^2} = 1 \text{ v.s. } H_1: \frac{\sigma_M^2}{\sigma_F^2} \neq 1), \alpha = 0.05$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{97161.9174}{577 - 4} \div 169,567$$

$$\hat{\sigma}_F^2 = 12,024^2 \div 144,577$$

$$F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169,567}{144,577} \div 1,173 \sim F_{573, 419} \quad \text{with } 1000 - 577 - 4$$

$$\text{since } F_{573, 419, 0.025} \div 0,8377 \text{ , and } F_{573, 419, 0,975} \div 1,1968$$

$$\Rightarrow \text{the rejection region is } \{F < 0,8377, F > 1,1968\}$$

$$\text{since } F^* = 1,173 < 1,1968 \text{ and } > 0,8377$$

$$\Rightarrow F^* \text{ isn't in the rejection region and we do not reject } H_0$$

(b) $H_0: \sigma_s^2 = \sigma_m^2$, $H_1: \sigma_s^2 \neq \sigma_m^2$, $\alpha = 0.05$ $N_m = 600$, $N_s = 400$

$$\hat{\sigma}_s^2 = \frac{SSR_s}{df_s} = \frac{56231.0382}{400-5} \doteq 142,357, \quad \hat{\sigma}_m^2 = \frac{SSE_m}{df_m} = \frac{100703.0471}{600-5} \doteq 169,249$$

$$F^* = \frac{\hat{\sigma}_m^2}{\hat{\sigma}_s^2} = \frac{169,249}{142,357} \doteq 1.189 \sim F_{595,395}$$

$$\text{since } F_{595,395,0.025} \doteq 0.8338, F_{595,395,0.975} \doteq 1.1954$$

$\Rightarrow$ the rejection region is $\{F > 1.1954, F < 0.8338\}$

$$\because F^* = 1.189 < 1.1954 \text{ and } > 0.8338$$

$\Rightarrow F^*$ isn't in the rejection region $\therefore$ we do not reject H_0

$$(C) NR^2 = 59.03 \sim \chi^2_4, \alpha = 0.05$$

$$\chi^2_{4,0.95} \doteq 9.488 \Rightarrow \text{rejection region is } \{NR^2 > 9.488\}$$

$\Rightarrow NR^2 = 59.03 > 9.488$ is in the rejection region $\Rightarrow$ 存在異質變異

$\rightarrow$ (b) 的 regression 中沒有 MARRIED $\therefore$ 無法解釋已婚, 未婚的誤差變異不同

$$(d) NR^2 = 78.82, \because \text{解釋變數共 } 4 + 4 + \underbrace{(2)}_{\text{degree of freedom}} = 14 \Rightarrow NR^2 \sim \chi^2_{14}$$

$$\text{and } \chi^2_{14,0.95} \doteq \underbrace{23.685}_{\text{critical value}} \Rightarrow \text{rejection region is } \{NR^2 > 23.685\}$$

$\Rightarrow NR^2 = 78.82 > 23.685$ is in the rejection region $\Rightarrow$ 存在異質變異

$$(e) \begin{cases} \text{EDUC, intercept} \\ \text{EXPER, METRO, FEMALE} \end{cases} \text{ 中 } \begin{cases} \text{robse} > \text{se} \Rightarrow \text{CI gotten wider} \\ \text{robse} < \text{se} \Rightarrow \text{CI gotten narrow} \end{cases}$$

$\rightarrow$ 沒有 inconsistency $\therefore$ use Robust SE is more accurater than SE

且 Robust SE may larger than SE

(f) 不衝突 { (b): 已婚 v.s, 未婚 SE 可能不同
(f): 已婚對WAGE影響可能不顯著

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

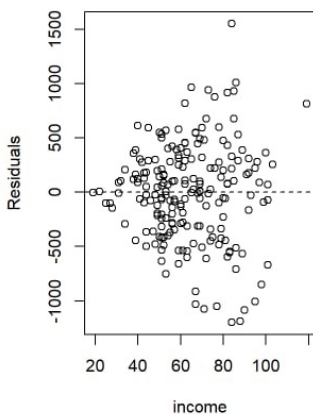
$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

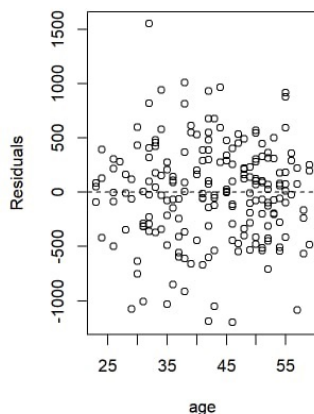
- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

8.16(a)
the 95% CI is : [-135.3298, -28.32302]

Residuals vs INCOME



Residuals vs AGE



① 中的變異隨

INCOME的±增加而±增加

① 有 heteroskedasticity

而②則沒有明顯的變化

c,

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## 8.16(c)  $\alpha = 0.05$   
## the  $F^*$  is : 3.104061  
## and  $F_c$  is : 1.428617
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$$H_0: \frac{\sigma_H^2}{\sigma_c^2} = 1 \quad \text{vs.} \quad H_1: \frac{\sigma_H^2}{\sigma_c^2} \neq 1$$

since $F^* > F_c \Rightarrow$ we reject H_0

d,

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## 8.16(d)  
## the 95% CI using heteroskedasticity robust standard errors is : [-139.323, -24.32986]
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(a): $[-135, 3298, -28.32302]$, (d) 的 CI 變寬

e,

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## 8.16(e)  
## the 95% CI using conventional GLS standard errors is : [-127.1453, -29.5814]  
## the 95% CI using robust conventional GLS standard errors is : [-127.7173, -29.00934]
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CI 的寬度: (d) > (a) > Robust GLS > GLS